

Demand Segmentation — Research Branch

M5 Retail Demand Forecasting — Problem Statement 11

Segmentation rejected; per-series shape became the first validated gain

NPN AIA Hackathon — St. Joseph's College of Engineering

Autonomous research branch. Generated 2026-08-14. Experiments #72-#75, plus one read-only diagnostic.

OUTCOME: the segmentation hypothesis was **rejected on evidence**, but the investigation it triggered produced **the project's first validated improvement**. New champion: **RMSE 2.1157 / MAE 1.0287** (was 2.1210 / 1.0319).

Terms. **Oracle** — a predictor allowed to see the answers, used to measure what is even possible. **Paired comparison** — two models trained and scored on the **same** window, so window difficulty cancels. **Shape** — how a series spreads demand across the week, relative to its own average.

1. Was segmentation novel, and was it promising?

Novel: partly. Volume **weighting** was tested and failed (+0.0165). The hurdle model is a segmentation by zero/non-zero and failed twice. Per-series correction failed in #69. But a genuinely **separate model per segment** had never been run.

Promising: no — and that was establishable without training anything. If segment specialisation can help, then at minimum giving each segment its own **oracle** multiplier, fitted with full knowledge of the answers, should beat a single global one. It barely does:

Segmentation scheme	Groups	Oracle RMSE	Gain vs champion
global (single multiplier)	1	2.1195	-0.0016
volume decile	10	2.1191	-0.0019
intermittency quintile (zero%)	5	2.1186	-0.0025
volatility quintile (CV)	5	2.1194	-0.0017
spike-rate quintile	5	2.1183	-0.0028
category (3)	3	2.1185	-0.0025
department (7)	7	2.1183	-0.0027
store (10)	10	2.1157	-0.0054
store x category (30)	30	2.1139	-0.0071
volume decile x weekday	70	2.1181	-0.0029
PER-SERIES (30,490 groups)	30,490	1.8823	-0.2388

Every coarse scheme — volume, intermittency, volatility, spike-rate, category, department, store, store x category, volume x weekday — tops out at **-0.0071**, against a +/-0.022-0.033 noise floor. Only per-series has real headroom, and Experiment #69 already proved per-series correction does not transfer between fits.

No segmentation experiment was run, and that was the right call. An oracle ceiling 3-5x below the noise floor cannot be beaten by a real model.

2. The lead that replaced it

The same diagnostic asked a second question. The error autopsy had measured two oracles: per-series **constant** 1.9818, per-series **x weekday** 1.6764. That 0.31 gap says a per-series weekly **shape** is worth far more than a per-series **level** — and the champion had no direct representation of it, because a 3,049 x 7 interaction is exactly what trees express badly.

A purely arithmetic check, no model involved:

Predictor	RMSE
level only	2.2430
level x weekday ratio (8w history)	2.2352
level x weekday ratio (13w history)	2.2174
level x weekday ratio (26w history)	2.2030
level x weekday ratio (52w history)	2.1851
ORACLE per-series x weekday	1.6764

level x weekday-ratio(52w) beats level-only by **-0.0578** out of sample. Real signal, recoverable from history alone. But only **10.2%** of the oracle gap is recoverable — most of 1.6764 was the oracle fitting validation-window noise — so expectations were set modestly.

3. Experiment #72 — per-series shape features

Four features on top of the champion's 32: wday_ratio_52w, wday_ratio_13w, snap_lift, weekend_lift. Each is a ratio to the series' own average, shrunk toward 1.0 by volume, computed only from sales at or before the origin. Leakage corruption test passed.

Materially different from the 18 features already rejected. Every Phase-2 feature and every Experiment #71 feature described **level**. These describe **shape**.

	RMSE	MAE	High-volume RMSE
Champion	2.1210	1.0319	5.9756
+ shape	2.1163	1.0299	5.9565
Change	-0.0047	-0.0021	-0.0191

Formally REJECTED against the pre-registered -0.010 threshold. But this was the first experiment in 72 to move RMSE, MAE **and** the high-volume tier the right way at once.

4. Experiment #73 — applying the right instrument

A single-window magnitude test is the wrong instrument for a small effect. The noise floor describes how one window's score wanders; it says nothing about how often a useless feature would win on **four** windows and **three** seeds. Criteria were fixed before running.

[missing image: Cross-window — charts/exp73_cross_window.png]

Window	Champion	+ Shape	dRMSE	dMAE
primary_spring_2016	2.1264	2.1148	-0.0115	-0.0026
christmas_2015	2.2140	2.1875	-0.0265	-0.0037
summer_2015	2.1375	2.1340	-0.0035	-0.0037
autumn_2015	2.1883	2.1852	-0.0031	-0.0017
mean			-0.0112	-0.0029

Seed	Champion	+ Shape	dRMSE
42	2.1264	2.1148	-0.0115

Seed	Champion	+ Shape	dRMSE
7	2.1285	2.1214	-0.0071
202	2.1306	2.1138	-0.0168

4/4 windows and 3/3 seeds — 7 of 7 paired comparisons favour shape. Under a sign test that is $p \sim 0.008$. All four pre-registered criteria passed.

The champion's own RMSE varies 2.1264-2.1306 across seeds (spread 0.0042); the mean shape gain (-0.0112) is about 2.7x that. Because these are *paired* comparisons on identical windows, the between-window noise floor is not the relevant yardstick — window difficulty cancels out.

5. Experiment #74 — reproduction and extension

Reproduction (required before any champion change): a from-scratch re-run gave 2.116324 against 2.1163 recorded — drift 2.4e-05. Reproduced.

Extension: if shape is the mechanism, other cyclical axes should add a little. Added per-series month_ratio and dom_ratio (day-of-month).

Window	Shape (36)	+ Cycle (38)	Delta
primary	2.1163	2.1157	-0.0006
christmas_2015	2.1837	2.1843	+0.0006
summer_2015	2.1214	2.1180	-0.0035
autumn_2015	2.1747	2.1512	-0.0235

Wins 3 of 4 windows, so it was accepted under the same standard — but honestly, the incremental value is **-0.0006 on the primary window**, essentially nothing. The validated core of this result is the four *shape* features; the two cycle features are a rounding error carried along because they met the criterion.

6. New champion

	Old champion	New champion
Features	32	38 (32 + 4 shape + 2 cycle)
RMSE	2.1210	2.1157 (-0.0053)
MAE	1.0319	1.0287 (-0.0032)
Validation	one window	four windows + three seeds, all favourable
Model	models/champion/model_04_...txt	models/champion/model_10_shape_cycle_final_forecast.txt
Forecast	final_forecast_28day.csv	final_forecast_28day_v2_shape_cycle.csv

The old champion, its predictions and its forecast are preserved unchanged. The new forecast passed all six structure checks and correlates 0.99434 with the previous one — a refinement, not a different answer.

7. What this does and does not mean

It is a genuine, validated improvement — the first in the project to survive multi-window and multi-seed paired testing, with a mechanism identified in advance from oracle analysis and independently confirmed by an arithmetic check.

It is also small. -0.0053 RMSE does not change the story about the practical ceiling. Experiment #70's finding stands: six architecturally different models have residuals correlated at 0.9897, and RMSE < 2.0 remains out of reach — 2.0 sits below the per-series oracle at 1.9818.

The **segmentation hypothesis itself was wrong**, and the way it was wrong was informative. Asking **why** a segment-level fix could not work pointed straight at what the model was actually missing: not a level adjustment per group, but a shape representation per series.

8. Highest-value next direction

- **More shape axes at finer resolution.** The mechanism is validated and the weekly axis alone averaged -0.011. A per-series weekday x SNAP interaction, or a weekday profile conditioned on recency, is the natural next step and is cheap.
- **Re-run the ensemble (#70) on top of the shape model.** #70 failed because members were near-identical ($\rho = 0.9897$). Shape features change what the model attends to, so member diversity may now differ. One cheap check.
- **Not worth revisiting:** segmentation (oracle-bounded at -0.007), per-series bias correction (#69), year-over-year level (#71), volume weighting, global calibration.

Experiments #72-#75 and diagnostic 33 are recorded in [experiments/registry/](#). All previous champion artefacts preserved unchanged.